Matrix Theory II–III: Rank and Systems of Linear Equations

Problem sheet for the Day 2 classes · CSIR NET / GATE / IIT JAM

Name: Date: Score: / 10

Instructions. 10 problems from CSIR NET, GATE (MA) and IIT JAM. A tag with an exam and year marks an actual previous-year question; a tag reading *Exam pattern* marks a problem written in the same style. MCQ: one correct option. MSQ: one or more correct options. NAT: write the number. Suggested time: 30 minutes. **The answer key is posted 24 hours after this sheet.**

1. JAM 2019 · NAT

If the system $x_1 + x_2 + x_3 + x_4 = 4,\ x_1 + 2x_2 + 3x_3 + 4x_4 = 5,\ x_1 + 3x_2 + 5x_3 + kx_4 = 5$ has no solution, then $k =$ _____.

2. CSIR NET Dec 2019 · MCQ

Let $A = \begin{pmatrix} 2 & 0 & 5 \\ 1 & 2 & 3 \\ -1 & 5 & 1 \end{pmatrix}$. The system $Ax = y$ has a solution

(A) only for $y = (x, 0, 0)^T,\ x \in \mathbb{R}$
(B) only for $y = (0, y, 0)^T,\ y \in \mathbb{R}$
(C) only for $y = (0, y, z)^T,\ y, z \in \mathbb{R}$
(D) for every $y \in \mathbb{R}^3$

3. JAM 2019 · MCQ

Let $T \in M_{m\times n}(\mathbb{R})$ and $V = \{X \in M_{n\times p}(\mathbb{R}) : TX = 0\}$. Then $\dim V$ is

(A) $pn - \operatorname{rank} T$
(B) $mn - p\operatorname{rank} T$
(C) $p(m - \operatorname{rank} T)$
(D) $p(n - \operatorname{rank} T)$

4. Exam pattern · MCQ

Let $n \geq 2$ and $A = (a_{ij})$ be the $n \times n$ matrix with $a_{ij} = i + j$. Then $\operatorname{rank} A$ is

(A) 1
(B) 2
(C) $n - 1$
(D) n

5. Exam pattern · MSQ

Let A be a 4×4 real matrix with $A^2 = 0$. Which of the following are true?

(A) $\operatorname{rank} A \leq 2$
(B) $\operatorname{rank} A = 2$ is possible
(C) $\operatorname{rank} A = 3$ is possible
(D) $I + A$ is invertible

6. Exam pattern · NAT

Let A be 5×7 of rank 4 and B be 7×6 of rank 5. The least possible value of $\operatorname{rank}(AB)$ is _____.

7. Exam pattern · MCQ

The system $x + y + z = 6,\ x + 2y + 3z = 10,\ x + 2y + \lambda z = \mu$ has infinitely many solutions when

(A) $\lambda = 3,\ \mu = 10$
(B) $\lambda = 3,\ \mu \neq 10$
(C) $\lambda \neq 3$
(D) $\lambda = 2,\ \mu = 10$

8. Exam pattern MSQ Let $A \in M_n(\mathbb{R})$, $n \geq 3$. Which of the following are true?

(A) if $\operatorname{rank} A = n - 1$, then $\operatorname{rank}(\operatorname{adj} A) = 1$

(B) if $\operatorname{rank} A \leq n - 2$, then $\operatorname{adj} A = 0$

(C) $\operatorname{rank}(\operatorname{adj} A) = n$ iff A is invertible

(D) $\operatorname{rank}(\operatorname{adj} A)$ can be 2

9. Exam pattern MSQ Let A be an $m \times n$ real matrix. Which of the following are true?

(A) if $\operatorname{rank} A = m$, then $Ax = b$ is consistent for every b

(B) if $\operatorname{rank} A = n$, then $Ax = b$ has at most one solution

(C) if $m < n$, then $Ax = 0$ has a non-zero solution

(D) if $m < n$, then $Ax = b$ is consistent for every b

10. Exam pattern NAT The rank of $\begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 5 \\ 3 & 4 & 5 & 6 \\ 4 & 5 & 6 & 7 \end{pmatrix}$ is _____.